Project Name: Voltage Protector Device

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1. Design Approach

My approach was to design a simple, fast-acting system focused on hardware safety. I built the firmware using a continuous state machine with non-blocking timers (using `millis()`) instead of basic delay functions. This ensures the microcontroller never freezes and is always monitoring the sensors. I also prioritized data safety by using the microcontroller's internal EEPROM memory, ensuring that any custom safety limits set by the user are permanently saved, even during a total power blackout.

2. Component Names and Working Principles

- **ATmega328P (Arduino Uno):** The main controller. It reads the analog sensor data, runs the fault logic, and handles serial communication with the PC.
- **Potentiometers (RV1 & RV2):** Used in the simulation to act as our incoming AC line voltage and load current sensors.
- **5V Relay & 2N2222 Transistor:** The relay is the physical switch that connects or disconnects the home appliance. Because the Arduino pins cannot provide enough current to switch the relay coil directly, the 2N2222 transistor acts as a current amplifier to drive it.
- **1N4007 Flyback Diode:** Placed in parallel across the relay coil. When the relay turns off, the coil releases a sharp magnetic voltage spike. This diode absorbs that spike so it doesn't burn out the transistor or reset the Arduino.
- **16x2 LCD & 4-Button Matrix:** Provides the local user interface. The buttons allow the user to interrupt the display, navigate a menu, and change safety limits without needing a computer.
- **Buzzer & LEDs:** Provide instant audio and visual alarms to tell the user if the grid is safe or if a fault occurred.

3. Testing and Results

I tested the system in Proteus under several simulated grid conditions. During normal operation (e.g., 180V and 8.8A), the relay stays closed, the green LED is on, and the LCD displays "NORMAL".

To test the fault logic, I dropped the voltage below the 180V limit, and later pushed the current past the 10A limit. In both tests, the system instantly disconnected the relay, sounded the buzzer, turned on the red LED, and displayed the specific fault (Under-Voltage or Over-Current) on the screen.

When returning the voltage to safe levels, the 5-second auto-recovery timer worked perfectly, delaying the reconnection. Finally, I tested the UART serial connection via the Virtual Terminal.

4. Discussion

One of the most important practical features I included is the 5-second recovery timer. In real life, power grids often fluctuate during a brownout. Forcing the system to wait 5 seconds before turning the appliance back on prevents rapid power cycling, which can destroy heavy loads like refrigerator compressors. Adding the flyback diode also ensures the physical circuit won't slowly destroy its own microcontroller over time. Overall, the combination of local button controls, PC serial communication, and physical hardware protections makes this a complete and reliable safety device.